KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4501 | 4 Credits: 4 | Program Core | Source-grounded

Fundamentals of Electronic

- To introduce the concept of electronic communication.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4501 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4501.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Electronics & Computer Engineering
Course code	4501
Course title	Fundamentals of Electronic
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:3 T:1 P:0) Periods per semester:60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To introduce the concept of electronic communication.
- ● To introduce the principle of amplitude modulation, frequency modulation and
- pulse modulation.
- ● To enrich on digital pulse modulation techniques, digital modulation schemes.
- ● To enrich on the data transmission methods and multiplexing techniques

Course outcomes

CO	Official outcome	Study level
CO1	Modulation (FM) techniques. 14 Understanding Explain the working of radio transmitters	See official syllabus
CO2	13 Understanding	See official syllabus
CO3	Explain various digital pulse modulation techniques 15 Understanding Summarize digital modulation schemes, multiplexing	See official syllabus
CO4	16 Understanding techniques and data transmission methods.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 2
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	techniques.	4	As specified
Module 2	M2.01 Explain the working of AM and FM transmitters 5 Understanding	3	As specified
Module 3	Explain various digital pulse modulation techniques	3	As specified
Module 4	Summarize digital modulation schemes, multiplexing techniques and data	3	As specified

MODULE 1

techniques.

This module is presented from the official syllabus scope for Fundamentals of Electronic. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: techniques.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain Communication systems. 3 Understanding. Interpret modulation parameters of

Explain Communication systems. 3 Understanding. Interpret modulation parameters of is an official topic in Module 1 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Communication systems. 3 Understanding. Interpret modulation parameters of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain communication systems. 3 understanding. interpret modulation parameters of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Communication systems. 3 Understanding. Interpret modulation parameters of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain Communication systems. 3 Understanding. Interpret modulation parameters of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

5 Understanding. different AM techniques Derive the expression for modulated

5 Understanding. different AM techniques Derive the expression for modulated is an official topic in Module 1 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding. different AM techniques Derive the expression for modulated using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding. different am techniques derive the expression for modulated should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding. different AM techniques Derive the expression for modulated means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 5 Understanding. different AM techniques Derive the expression for modulated □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding. signals and frequency spectrum of FM Solve for modulation index, bandwidth and

3 Understanding. signals and frequency spectrum of FM Solve for modulation index, bandwidth and is an official topic in Module 1 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. signals and frequency spectrum of FM Solve for modulation index, bandwidth and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. signals and frequency spectrum of fm solve for modulation index, bandwidth and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding. signals and frequency spectrum of FM Solve for modulation index, bandwidth and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 3 Understanding. signals and frequency spectrum of FM Solve for modulation index, bandwidth and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

power for various modulation schemes for 3 Applying given signal parameters Concept of modulation. Introduction to communication system: Elements of communication systems, Examples of analog communication systems, Frequency bands, Need for modulation. Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB. Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM. Comparison of AM and FM, Applications of AM and FM. Explain the working of radio transmitters and receivers

power for various modulation schemes for 3 Applying given signal parameters Concept of modulation. Introduction to communication system: Elements of communication systems, Examples of analog communication systems, Frequency bands, Need for modulation. Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB. Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM. Comparison of AM and FM, Applications of AM and FM. Explain the working of radio transmitters and receivers is an official topic in Module 1 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify power for various modulation schemes for 3 Applying given signal parameters Concept of modulation. Introduction to communication system: Elements of communication systems, Examples of analog communication systems, Frequency bands, Need for modulation. Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB. Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM. Comparison of AM and FM, Applications of AM and FM. Explain the working of radio transmitters and receivers using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, power for various modulation schemes for 3 applying given signal parameters concept of modulation. introduction to communication system: elements of communication systems, examples of analog communication systems, frequency bands, need for modulation. amplitude modulation: definition, expression of am wave, modulation index, am wave representation, bandwidth and power calculation, frequency spectrum of am, dsbsc, ssb and vsb. frequency modulation: definition, mathematical representation of fm, modulation index, frequency spectrum of fm. comparison of am and fm, applications of am and fm. explain the working of radio transmitters and receivers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What power for various modulation schemes for 3 Applying given signal parameters Concept of modulation. Introduction to communication system: Elements of communication systems, Examples of analog communication systems, Frequency bands, Need for modulation. Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB. Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM. Comparison of AM and FM, Applications of AM and FM. Explain the working of radio transmitters and receivers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ power for various modulation schemes for 3 Applying given signal parameters Concept of modulation. Introduction to communication system: Elements of communication systems, Examples of analog communication systems, Frequency bands, Need for modulation. Amplitude

Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB. Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM. Comparison of AM and FM, Applications of AM and FM. Explain the working of radio transmitters and receivers ▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition/meaning ▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯, steps, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Technical terms English-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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techniques.

  M1.01     Explain Communication systems.                     3
Understanding.
            Interpret modulation parameters of
  M1.02                                                         5
Understanding.
            different AM techniques
            Derive the expression for modulated
  M1.03                                                         3
Understanding.
            signals and frequency spectrum of FM
            Solve for modulation index, bandwidth and
  M1.04     power for various modulation schemes for            3
Applying
            given signal parameters

Contents:
Concept of modulation.
Introduction to communication system: Elements of communication systems, Examples
of analog communication systems, Frequency bands, Need for modulation.
Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM
wave
representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC,
SSB
and VSB.
Frequency Modulation: Definition, Mathematical representation of FM, Modulation
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

M2.01 Explain the working of AM and FM transmitters 5 Understanding

This module is presented from the official syllabus scope for Fundamentals of Electronic. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: M2.01 Explain the working of AM and FM transmitters 5 Understanding
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain the working of AM and FM transmitters 5 Understanding Explain the working of AM and FM receivers

Explain the working of AM and FM transmitters 5 Understanding Explain the working of AM and FM receivers is an official topic in Module 2 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the working of AM and FM transmitters 5 Understanding Explain the working of AM and FM receivers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the working of am and fm transmitters 5 understanding explain the working of am and fm receivers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the working of AM and FM transmitters 5 Understanding Explain the working of AM and FM receivers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain the working of AM and FM transmitters 5 Understanding Explain the working of AM and FM receivers [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

5 Understanding Compare various analog pulse modulation

5 Understanding Compare various analog pulse modulation is an official topic in Module 2 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Compare various analog pulse modulation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding compare various analog pulse modulation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Compare various analog pulse modulation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Understanding Compare various analog pulse modulation □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding. techniques. AM transmitter: AM Transmitter - low level and high level, AM collector modulator, Balanced modulator FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis circuits Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. Superheterodyne receiver, AM receiver(block diagram only), choice of IF in receivers , FM receiver(block diagram only), comparison of FM and AM receiver AM demodulator circuit: diode detector. Pulse Modulation Schemes: PAM, PWM, PPM, PCM

3 Understanding. techniques. AM transmitter: AM Transmitter - low level and high level, AM collector modulator, Balanced modulator FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis circuits Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. Superheterodyne receiver, AM receiver(block diagram only), choice of IF in receivers , FM receiver(block diagram only), comparison of FM and AM receiver AM demodulator circuit: diode detector. Pulse Modulation Schemes: PAM, PWM, PPM, PCM is an official topic in Module 2 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. techniques. AM transmitter: AM Transmitter - low level and high level, AM collector modulator, Balanced modulator FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis circuits Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. Superheterodyne receiver, AM receiver(block diagram only), choice of IF in receivers , FM receiver(block diagram only), comparison of FM and AM receiver AM demodulator circuit: diode detector. Pulse Modulation Schemes: PAM, PWM, PPM, PCM using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. techniques. am transmitter: am transmitter - low level and high level, am collector modulator, balanced modulator fm transmitter: fm transmitter(armstrong method), pre-emphasis and de-emphasis circuits characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. superheterodyne receiver, am receiver(block diagram only), choice of if in receivers , fm receiver(block diagram only), comparison of fm and am receiver am demodulator circuit: diode detector. pulse modulation schemes: pam, pwm, ppm, pcm should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding. techniques. AM transmitter: AM Transmitter - low level and high level, AM collector modulator, Balanced modulator FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis circuits Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. Superheterodyne receiver, AM receiver(block diagram only), choice of IF in receivers , FM receiver(block diagram only), comparison of FM and AM receiver AM demodulator circuit: diode detector. Pulse Modulation Schemes: PAM, PWM, PPM, PCM means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding. techniques. AM transmitter: AM Transmitter - low level and high level, AM collector modulator, Balanced modulator FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis circuits Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. Superheterodyne receiver, AM receiver(block diagram only), choice of IF in receivers , FM receiver(block diagram only), comparison of FM and AM receiver AM demodulator circuit: diode detector. Pulse Modulation Schemes: PAM, PWM, PPM, PCM □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
M2.01     Explain the working of AM and FM transmitters           5
Understanding

            Explain the working of AM and FM receivers
  M2.02                                                             5
Understanding

            Compare various analog pulse modulation
  M2.03                                                             3
Understanding.
            techniques.
           Series Test – I                                          1

Contents:
AM transmitter: AM Transmitter - low level and high level, AM collector modulator,
Balanced modulator
FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis
circuits
Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise
figure.
Superheterodyne receiver, AM receiver(block diagram only), choice of IF in
receivers , FM
receiver(block diagram only), comparison of FM and AM receiver
AM demodulator circuit: diode detector.
Pulse Modulation Schemes: PAM, PWM, PPM, PCM
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain various digital pulse modulation techniques

This module is presented from the official syllabus scope for Fundamentals of Electronic. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain various digital pulse modulation techniques
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

1 Remembering

1 Remembering is an official topic in Module 3 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Remembering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 remembering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Remembering means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 1 Remembering [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate Sampling and Quantization. 4 Understanding Interpret the PCM systems

Illustrate Sampling and Quantization. 4 Understanding Interpret the PCM systems is an official topic in Module 3 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Sampling and Quantization. 4 Understanding Interpret the PCM systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate sampling and quantization. 4 understanding interpret the pcm systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate Sampling and Quantization. 4 Understanding Interpret the PCM systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Illustrate Sampling and Quantization. 4 Understanding Interpret the PCM systems □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding Explain DPCM, DM and ADM M1.04 6 Understanding Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system, Sampling and Quantization: Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error. PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation (Block diagram explanation only)

4 Understanding Explain DPCM, DM and ADM M1.04 6 Understanding Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system, Sampling and Quantization: Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error. PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation (Block diagram explanation only) is an official topic in Module 3 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Explain DPCM, DM and ADM M1.04 6 Understanding Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system, Sampling and Quantization: Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error. PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation (Block diagram explanation only) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding explain dpcm, dm and adm m1.04 6 understanding necessity of digital communication systems, block diagram and sub-system description of a digital communication system, sampling and quantization: definition of sampling, types of sampling techniques, sampling theorem and nyquist rate, definition of quantization, uniform and non-uniform quantization, quantization error. pcm, dpcm, delta modulation, noise in delta modulation: slope overload and granular noise, adaptive delta modulation (block diagram explanation only) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Explain DPCM, DM and ADM M1.04 6 Understanding Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system, Sampling and Quantization: Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error. PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation (Block diagram explanation only) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding Explain DPCM, DM and ADM M1.04 6 Understanding Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system, Sampling and Quantization: Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error. PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation (Block diagram explanation only) □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain various digital pulse modulation techniques
            Define digital communication systems
  M3.01                                                          1
Remembering

  M3.02     Illustrate Sampling and Quantization.                4
Understanding
            Interpret the PCM systems
  M3.03                                                          4
Understanding

            Explain DPCM, DM and ADM
  M1.04                                                          6
Understanding

Contents:
Necessity of digital communication systems, Block diagram and sub-system
description of a
digital communication system,
Sampling and Quantization: Definition of sampling, Types of sampling techniques,
Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-
uniform
Quantization, Quantization error.
PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular
noise, Adaptive Delta modulation (Block diagram explanation only)
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Summarize digital modulation schemes, multiplexing techniques and data

This module is presented from the official syllabus scope for Fundamentals of Electronic. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize digital modulation schemes, multiplexing techniques and data
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Summarize digital modulation schemes.

Summarize digital modulation schemes. is an official topic in Module 4 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize digital modulation schemes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize digital modulation schemes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize digital modulation schemes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Summarize digital modulation schemes. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Compare various multiplexing techniques

Compare various multiplexing techniques is an official topic in Module 4 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare various multiplexing techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare various multiplexing techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare various multiplexing techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Compare various multiplexing techniques □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Describe data transmission methods 4 Understanding Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, QPSK Multiplexing Techniques: TDM & FDM-Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. Data transmission methods: Simplex, Half-duplex, Full duplex, Synchronous and Asynchronous Text/Reference: T/R Book Title/Author T1 Electronic Communication Systems – George Kennedy – TMH T2 Communication Systems - Simon Haykin R1 Communication Systems(Analog and Digital) – Sanjay Sharma R2 Electronic Communication Systems - Wayne Thomasi. R3 Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014 R4 Digital Communications – Sanjay Sharma Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117/105/117105143/ 2 https://nptel.ac.in/courses/117101051/

Describe data transmission methods 4 Understanding Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, QPSK Multiplexing Techniques: TDM & FDM-Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. Data transmission methods: Simplex, Half-duplex, Full duplex, Synchronous and Asynchronous Text/Reference: T/R Book Title/Author T1 Electronic Communication Systems – George Kennedy – TMH T2 Communication Systems - Simon Haykin R1 Communication Systems(Analog and Digital) – Sanjay Sharma R2 Electronic Communication Systems - Wayne Thomasi. R3 Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014 R4 Digital Communications – Sanjay Sharma Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117/105/117105143/ 2 https://nptel.ac.in/courses/117101051/ is an official topic in Module 4 of Fundamentals of Electronic. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe data transmission methods 4 Understanding Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, QPSK Multiplexing Techniques: TDM & FDM-

Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. Data transmission methods: Simplex, Half-duplex, Full duplex, Synchronous and Asynchronous Text/Reference: T/R Book Title/Author T1 Electronic Communication Systems – George Kennedy – TMH T2 Communication Systems - Simon Haykin R1 Communication Systems(Analog and Digital) – Sanjay Sharma R2 Electronic Communication Systems - Wayne Thomasi. R3 Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014 R4 Digital Communications – Sanjay Sharma Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117/105/117105143/ 2 https://nptel.ac.in/courses/117101051/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe data transmission methods 4 understanding digital modulation schemes generation and reception of ask, bfsk, bpsk, comparison of ask, bfsk and bpsk. concept of dpsk, qpsk multiplexing techniques: tdm & fdm-concept, block diagram, advantages, disadvantages, applications, and comparison. data transmission methods: simplex, half-duplex, full duplex, synchronous and asynchronous text/reference: t/r book title/author t1 electronic communication systems – george kennedy – tmh t2 communication systems - simon haykin r1 communication systems(analog and digital) – sanjay sharma r2 electronic communication systems - wayne thomasi. r3 digital communication -john g. proakis, masoud salehi, , mcgraw hill education edition, 2014 r4 digital communications – sanjay sharma online resources: sl.no website link 1 https://nptel.ac.in/courses/117/105/117105143/ 2 https://nptel.ac.in/courses/117101051/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe data transmission methods 4 Understanding Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, QPSK Multiplexing Techniques: TDM & FDM-Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. Data transmission methods: Simplex, Half-duplex, Full duplex, Synchronous and Asynchronous Text/Reference: T/R Book Title/Author T1 Electronic Communication Systems – George Kennedy – TMH T2 Communication Systems - Simon Haykin R1 Communication Systems(Analog and Digital) – Sanjay Sharma R2 Electronic Communication Systems - Wayne Thomasi. R3 Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014 R4 Digital Communications – Sanjay Sharma Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117/105/117105143/ 2 https://nptel.ac.in/courses/117101051/ means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Describe data transmission methods 4 Understanding Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, QPSK Multiplexing Techniques: TDM & FDM-Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. Data transmission methods: Simplex, Half-duplex, Full duplex, Synchronous and Asynchronous Text/Reference: T/R Book Title/Author T1 Electronic Communication Systems – George Kennedy – TMH T2 Communication Systems - Simon Haykin R1 Communication Systems(Analog and Digital) – Sanjay Sharma R2 Electronic Communication Systems - Wayne Thomasi. R3 Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014 R4 Digital Communications – Sanjay Sharma Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117/105/117105143/ 2 https://nptel.ac.in/courses/117101051/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize digital modulation schemes, multiplexing techniques and data
           transmission methods.
  M4.01    Summarize digital modulation schemes.              8
Understanding

  M4.02    Compare various multiplexing techniques            4
Understanding

  M4.03    Describe data transmission methods                 4
Understanding

           Series Test – II                                   1

Contents:
Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK,
Comparison
of ASK, BFSK and BPSK.
Concept of DPSK,
QPSK Multiplexing Techniques: TDM & FDM-Concept, Block diagram, Advantages,
Disadvantages, Applications, and Comparison.
Data transmission methods: Simplex, Half-duplex, Full duplex, Synchronous and
Asynchronous

Text/Reference:

  T/R                                   Book Title/Author
   T1     Electronic Communication Systems – George Kennedy – TMH
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain Communication systems. 3 Understanding. Interpret modulation parameters of. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Communication systems. 3 Understanding. Interpret modulation parameters of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding. different AM techniques Derive the expression for modulated. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding. different AM techniques Derive the expression for modulated*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding. signals and frequency spectrum of FM Solve for modulation index, bandwidth and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding. signals and frequency spectrum of FM Solve for modulation index, bandwidth and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain power for various modulation schemes for 3 Applying given signal parameters Concept of modulation. Introduction to communication system: Elements of communication systems, Examples of analog communication systems, Frequency bands, Need for modulation. Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB. Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM. Comparison of AM and FM, Applications of AM and FM. Explain the working of radio transmitters and receivers. (3 marks)

Answer: Begin with the standard definition or identity of *power for various modulation schemes for 3 Applying given signal parameters Concept of modulation. Introduction to communication system: Elements of communication systems, Examples of analog communication systems, Frequency bands, Need for modulation. Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB. Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM. Comparison of AM and FM, Applications of AM and FM. Explain the working of radio transmitters and receivers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Explain the working of AM and FM transmitters 5 Understanding Explain the working of AM and FM receivers. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the working of AM and FM transmitters 5 Understanding Explain the working of AM and FM receivers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding Compare various analog pulse modulation. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding Compare various analog pulse modulation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding. techniques. AM transmitter: AM Transmitter - low level and high level, AM collector modulator, Balanced modulator FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis circuits Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. Superheterodyne receiver, AM receiver(block diagram only), choice of IF in receivers , FM receiver(block diagram only), comparison of FM and AM receiver AM demodulator circuit: diode detector. Pulse Modulation Schemes: PAM, PWM, PPM, PCM. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding. techniques. AM transmitter: AM Transmitter - low level and high level, AM collector modulator, Balanced modulator FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis circuits Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. Superheterodyne receiver, AM receiver(block diagram only), choice of IF in receivers , FM receiver(block diagram only), comparison of FM and AM receiver AM demodulator circuit: diode detector. Pulse Modulation Schemes: PAM, PWM, PPM, PCM*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 1 Remembering. (3 marks)

Answer: Begin with the standard definition or identity of *1 Remembering*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate Sampling and Quantization. 4 Understanding Interpret the PCM systems. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate Sampling and Quantization. 4 Understanding Interpret the PCM systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding Explain DPCM, DM and ADM M1.04 6 Understanding Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system, Sampling and Quantization: Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error. PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation (Block diagram explanation only). (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding Explain DPCM, DM and ADM M1.04 6 Understanding Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system, Sampling and Quantization: Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error. PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation (Block diagram explanation only)*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Summarize digital modulation schemes.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize digital modulation schemes.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Compare various multiplexing techniques. (3 marks)

Answer: Begin with the standard definition or identity of *Compare various multiplexing techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe data transmission methods 4 Understanding Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, QPSK Multiplexing Techniques: TDM & FDM-Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. Data transmission methods: Simplex, Half-duplex, Full duplex, Synchronous and Asynchronous Text/Reference: T/R Book Title/Author T1 Electronic Communication Systems – George Kennedy – TMH T2 Communication Systems - Simon Haykin R1 Communication Systems(Analog and Digital) – Sanjay Sharma R2 Electronic Communication Systems - Wayne Thomasi. R3 Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014 R4 Digital Communications – Sanjay Sharma Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117/105/117105143/ 2 https://nptel.ac.in/courses/117101051/. (3 marks)

Answer: Begin with the standard definition or identity of *Describe data transmission methods 4 Understanding Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, QPSK Multiplexing Techniques: TDM & FDM-Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. Data transmission methods: Simplex,* *Half-duplex, Full duplex, Synchronous and Asynchronous Text/Reference: T/R Book Title/Author T1 Electronic Communication Systems – George Kennedy – TMH T2 Communication Systems - Simon Haykin R1 Communication Systems(Analog and Digital) – Sanjay Sharma R2 Electronic Communication Systems - Wayne Thomasi. R3 Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014 R4 Digital Communications – Sanjay Sharma Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117/105/117105143/ 2 https://nptel.ac.in/courses/117101051/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain Communication systems. 3 Understanding. Interpret modulation parameters of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain the working of AM and FM transmitters 5 Understanding Explain the working of AM and FM receivers**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **1 Remembering**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Summarize digital modulation schemes.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.

- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://nptel.ac.in/courses/117/105/117105143/ https://nptel.ac.in/courses/117101051/

IN-PAGE SEARCH

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